

Duc Hoang

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SUMMARY

PhD candidate in High-Energy Physics at MIT and CERN developing real-time machine learning systems for nanosecond-scale decision making. My work spans FPGA-optimized neural architectures, adaptive hardware-native learning algorithms, and large-scale statistical analyses for precision Higgs measurements.

EDUCATION

- **Massachusetts Institute of Technology (MIT)** 2021 - Present
PhD, Physics. Thesis: Harder, Better, Faster, Stronger: Pushing the Data Frontier at the Large Hadron Collider Cambridge, USA
- **Rhodes College** 2017-2021
B.S., Physics, Summa Cum Laude. Thesis: Triggering on electron-nucleon scattering events at LDMX Memphis, USA

PROFESSIONAL EXPERIENCE

- **MIT & European Organization for Nuclear Research (CERN)** 2021 - Present
Graduate Researcher & Next Generation Trigger PhD Fellow Cambridge, USA & Geneva, Switzerland
 - Coordinated multi-institutional team to develop **ultra-low-latency FPGA-accelerated ML models** for CMS Level-1 Trigger, which processes approximately 10% of global internet traffic in real time. [Hear me explain it along the trigger leadership team](#) [2]
 - Built **first FPGA-based ML algorithms for real-time jet tagging** and delivered cross-platform, low-latency data transfer tests between different FPGA board platforms. [3]
 - Analyzed all Run 2 CMS data (a few PBs) to deliver **most sensitive Higgs $\rightarrow b\bar{b}$ measurement in the V(qq)H(bb) decay channel**, invited presentation at Rencontres de Moriond 2025. [4]
 - Supervised MIT undergraduates and masters on research projects involving the CMS Level-1 Trigger system and the development of **novel machine learning architectures for ultra-fast and efficient inference on FPGAs**. [1,9].
 - Designed and led practical CERN training sessions on machine learning techniques for FPGA-based Level-1 Trigger applications for CERN students and researchers.
 - Developed data analysis projects for MIT 8.16/8.316: Data Science in Physics, which is published on MITx course [Computational Data Science in Physics](#).
- **Espresso (MIT Sandbox Team)** 2025 - Present
Co-founder Cambridge, USA
 - Developing **efficient real-time and local data processing capabilities** for space and defense industry.
 - Defined the company's technical vision and led early customer discoveries focusing on space applications.
- **Fermi National Accelerator Laboratory** 2019 - 2021
Undergraduate Researcher Chicago, USA
 - Developer of **hls4ml**, A package for **ultra-low-latency machine learning inference in FPGAs** using high level synthesis language (HLS). [5]
 - Designed superconducting magnet quench detection systems [6]
 - Analyzed simulation data to predict potential signal gains for the real-time trigger systems for Light Dark Matter Experiment (LDMX). [7]
 - Machine Learning analysis to infer hyper-parameter optimization performance of models used in the MINERvA Experiment [8].

SELECTED PUBLICATIONS & SOFTWARE

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| <p>[1] D. Hoang et al. "KANELE: Kolmogorov-Arnold Networks for Efficient LUT-based Evaluation," Proceedings of the 2026 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA'26), Seaside, CA, USA, 2026. https://arxiv.org/abs/2512.12850</p> <p>[2] CMS Collaboration, "The Phase-2 Upgrade of the CMS Level-1 Trigger", https://cds.cern.ch/record/2714892</p> <p>[3] CMS Collaboration, 2025. "Multiclass object tagging in the CMS Phase-2 Level-1 Correlator Trigger." CMS Detector Performance Summary. https://cds.cern.ch/record/2936315.</p> <p>[4] CMS Collaboration, "Search for a boosted Higgs</p> | <p>boson decaying to bottom quark pairs in association with a hadronically decaying W or Z boson with the CMS detector using proton-proton collisions at $\sqrt{s} = 13$ TeV", https://cds.cern.ch/record/2928061</p> <p>[5] hls4ml Community, "hls4ml: An Open-Source Codesign Workflow to Empower Scientific Low-Power Machine Learning Devices"</p> <p>[6] D. Hoang et al., "IntelliQuench: an adaptive machine learning system for detection of superconducting magnet quenches," in <i>IEEE Transactions on Applied Superconductivity</i>, doi: 10.1109/TASC.2021.3058229.</p> |
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- [7] **LDMX Collaboration**, "Current Status and Future Prospects for the Light Dark Matter eXperiment" [[arXiv:2203.08192\[hep-ex\]](#)]. (ICMLA), Boca Raton, FL, USA, 2019, pp. 1388-1391. DOI: [10.1109/ICMLA.2019.00226](#)
- [8] **D. Hoang et al.** "Inferring Convolutional Neural Networks' Accuracies from Their Architectural Characterizations," *2019 18th IEEE International Conference On Machine Learning And Applications*
- [9] **CMS Collaboration**. 2024. "NNPuppiTaus: PUPPI Tau Reconstruction in the Level-1 Trigger with Real-Time Machine Learning." CMS Detector Performance Summary. <https://cds.cern.ch/record/2895660>.

MENTORSHIP

Aarush Gupta, MIT BS/MS Thesis (CS & Physics). *Subsequent position: Jane Street*
Orion Foo, MIT BS (CS & Physics).

HONORS AND AWARDS

- **The 2025 Breakthrough Prize in Fundamental Physics** 2025
Received as a member of the CMS Collaboration 
 ◦ The prize is awarded to physicists from theoretical, mathematical, or experimental physics that have made transformative contributions to fundamental physics, and specifically for recent advances.
- **2025 European Physical Society Conference on High Energy Physics (EPS-HEP) Poster Prize** 2025
European Physical Society (EPS) 
 ◦ Awarded for best poster at EPS-HEP 2025 (out of more than 100 posters), a highly prestigious biannual international conference featuring major results from leading high-energy physics experiments and collaborations.
- **MIT Public Service Fellow (PKG) Fellowship** 2025
Priscilla King Gray Public Service Center - MIT 
 ◦ Founded and organized, along with other researchers from Central Vietnam, a STEM [summer school](#), providing under-privileged high school students in Central Vietnam the opportunity to study emerging fields such as artificial intelligence, data science, programming, renewable energy, and quantum technology at no cost.
- **Next Generation Trigger PhD fellow.** 2024
CERN 
 ◦ Awarded for exceptional PhD students to work along CERN scientists to develop the next generation trigger system at the Large Hadron Collider.
- **2024-25 Christiaan Huygens Graduate Fellowship - MIT.** 2024
MIT School of Science
 ◦ Fellowship providing financial support for the full academic year in the School of Science for students with non-traditional backgrounds.
- **MIT-The Royal Game of Ur Champion** 2023
MIT Playur II Tournament 
 ◦ The Royal Game of Ur is the oldest board game in the world.
- **Henry W Kendall (1955) first year graduate fellowship.** 2021
MIT Laboratory for Nuclear Science
 ◦ Financial support for first year graduate students.
- **Undergraduate Awards.** 2017-2021
Rhodes College
 ◦ Research Award in Physics, Council on Undergraduate Research (CUR) Physics & Astronomy Student Travel Award, W.O Shewmaker Award for Excellence in First-Year Search Program, Award for Excellence in Physics by a First-Year Student, Summa Cum Laude, Sigma Pi Sigma, Phi Beta Kappa, Physics Honors.